Model Architecture: Llama-3.1-8B-Instruct

Generated by sg_modelload

October 24, 2026

Contents

1 Model Overview **2**

2 Mathematical Representation **2**
- 2.1 The model as a composition of layers . . . 2
- 2.2 The layers, top to bottom . . . 3
- 2.3 Neurons and attention heads . . . 4
- 2.4 Architecture and parameters . . . 4
- 2.5 Output representations . . . 4
- 2.6 Intermediate states and sub-network decomposition . . . 5
- 2.7 Local linear representation . . . 5
- 2.8 Summary of notation . . . 5

3 Forward Pass in Pseudocode **6**

4 Compact Architecture (repeated layers written once) **6**
- 4.1 Modules outside the repeated stacks . . . 6
- 4.2 Language-model decoder: `model.layers` . . . 6

5 What Each Layer Does **7**

6 Parameter Count **7**

A Full Layer-by-Layer Table **8**

B Per-Layer Parameter Count **14**

1 Model Overview

Model name	Llama-3.1-8B-Instruct
Transformers class	`LlamaForCausalLM`
Path	`/models/Llama-3.1-8B-Instruct`
Total parameters	8,030,261,248 (8.03B)
Weight dtype	bfloat16
Hidden size d	4,096
Decoder layers L	32
Query heads H	32
Key/value heads H_{kv}	8
Head dim d_h	128
MLP width d_{ff}	14,336
Vocabulary V	128,256
Max context length	131,072
RMSNorm ϵ	1e-05
Tied input/output embeddings	False
Activation	SiLU
RoPE base θ	500,000

How to read this document. Section 2 treats the model mathematically, top-down: first the whole network as one composition $\Psi \circ \mathcal{B}^{(L)} \circ \cdots \circ \mathcal{B}^{(1)} \circ E$ of layers, then a definition of every layer down to single neurons and attention heads, then parameters, outputs, intermediate states and the local linear response. Section 3 gives readable pseudocode for the forward pass and for every distinct block type. Section 4 lists the weights of each block type *once*, with how many times it repeats. Section 5 explains in plain language what every kind of layer does. The original layer-by-layer table with every repetition written out is in Appendix A.

Basic notation

B, T	batch size, sequence length (tokens)
d	hidden size (4096)
V	vocabulary size (128,256)
H, H_{kv}, d_h	query heads, key/value heads, size of one head
W_*, b_*	learned weight matrix / bias of the named projection
$\odot$, σ	element-wise product, logistic sigmoid
$f^{\circ n}$	f applied n times in a row ($f \circ f \circ \cdots \circ f$)
$g \circ f$	first f, then g (composition reads right to left)
RMSNorm(x)	$\gamma \odot x / \sqrt{\text{mean}(x^2) + \epsilon}$ (learned scale γ)
LN(x)	LayerNorm: $\gamma \odot (x - \mu)/\sigma + \beta$
RoPE	rotary position embedding (rotates query/key pairs by an angle set by position)

2 Mathematical Representation

We describe the model *top-down*. First the whole network is written as one composition of functions (Section 2.1). Then every function that appears in that composition is defined (Section 2.2), down to individual neurons and attention heads. The remaining subsections describe the parameters, the outputs, the intermediate states, and the local linear behaviour.

2.1 The model as a composition of layers

Definition 2.1 (Llama-3.1-8B-Instruct as an L-layer decoder)**.** The model is a function $f_\theta : \mathcal{X} \to \mathcal{Y}$ with parameters θ. It maps a sequence of T token ids $x \in \mathcal{X} = \{1, \ldots, V\}^T$ to one logit vector per position, $y \in \mathcal{Y} = \mathbb{R}^{T \times V}$ ($V = 128{,}256$). It is a composition of an embedding E, $L = 32$ decoder blocks $\mathcal{B}^{(\ell)}$ and an output head Ψ:

$$f_\theta \;=\; \Psi \circ \mathcal{B}^{(L)} \circ \mathcal{B}^{(L-1)} \circ \cdots \circ \mathcal{B}^{(2)} \circ \mathcal{B}^{(1)} \circ E \tag{1}$$

Every block maps $\mathbb{R}^{T \times d} \to \mathbb{R}^{T \times d}$ ($d = 4096$), so the blocks can be stacked in any number. Block ℓ has its own parameters θ_ℓ and is stored in the checkpoint as `model.layers.`$(\ell - 1)$.

Definition 2.2 (Block types)**.** All L blocks have the same functional form $\mathcal{B}_{\mathrm{D}}$ (Decoder block). They differ only in their weights: $\mathcal{B}^{(\ell)} = \mathcal{B}_{\mathrm{D}}(\,\cdot\,;\theta_\ell)$. Written out, with the type as a subscript:

$$f_\theta = \Psi \circ \mathcal{B}_{\mathrm{D}}^{(32)} \circ \cdots \circ \mathcal{B}_{\mathrm{D}}^{(3)} \circ \mathcal{B}_{\mathrm{D}}^{(2)} \circ \mathcal{B}_{\mathrm{D}}^{(1)} \circ E \tag{2}$$

Grouping the repetitions with composition powers ($\mathcal{B}^{\circ n} = n$ consecutive blocks of the same type, each with its own weights) gives the compact form

$$f_\theta = \Psi \circ \left(\mathcal{B}_{\mathrm{D}}^{\circ 32}\right) \circ E \tag{3}$$

Layer pattern.

Symbol	Block type	Count	Params / block	Checkpoint indices
$\mathcal{B}_{\mathrm{D}}$	Decoder block	32	218,112,000	0–31

Language-model decoder (32 layers). layers 0–31: $[\mathcal{B}_{\mathrm{D}}]$ repeated 32× (checkpoint indices, i.e. $\ell - 1$).

2.2 The layers, top to bottom

Every function in the composition above is defined here. Only the attention (or linear-attention) sub-layer mixes information *between* positions t. Every other sub-layer (norms, feed-forward networks, projections) acts on each token vector $h_t \in \mathbb{R}^d$ independently, e.g. $\mathrm{FFN}(h)_t = \mathrm{FFN}(h_t)$. All weights follow the convention $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, so a projection of a token vector is Wu (as in an affine layer $f_i(u) = A^{(i)}u + b^{(i)}$).

Definition 2.3 (Token embedding)**.** $E : \{1, \ldots, V\}^T \to \mathbb{R}^{T \times d}$ looks up one row of the embedding matrix $W_E \in \mathbb{R}^{V \times d}$ per token:

$$E(x)_t = W_E[x_t, :], \qquad t = 1, \ldots, T. \tag{4}$$

Definition 2.4 (Normalisation)**.** $\mathrm{RMSNorm}(u) = \gamma \odot u / \sqrt{\frac{1}{d}\sum_i u_i^2 + \epsilon}$, with a learned scale $\gamma \in \mathbb{R}^d$, applied to each token vector separately. It fixes the scale of a vector without changing its direction. Every norm in the model has its own γ, distinguished by a subscript (e.g. $\mathrm{RMSNorm}_1$, $\mathrm{RMSNorm}_{\text{final}}$).

Definition 2.5 (Block $\mathcal{B}_{\mathrm{D}}$: Decoder block)**.** A decoder block of this type maps $h \in \mathbb{R}^{T \times 4096}$ to the same space. It is a composition of 2 residual sub-layers, each of the form $(\mathrm{id} + g)(h) = h + g(h)$:

$$\mathcal{B}_{\mathrm{D}} = (\mathrm{id} + \mathrm{FFN} \circ \mathrm{RMSNorm}_2) \circ (\mathrm{id} + \mathrm{Attn} \circ \mathrm{RMSNorm}_1) \tag{5}$$

Step by step, the sub-layers are:

$$\begin{aligned}
a &= h + \mathrm{Attn}(\mathrm{RMSNorm}_1(h)) && \text{pre-norm + residual}\\
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{RMSNorm}_2(a)) && \text{pre-norm + residual}\\
Q &= W_Q\, u && H = 32 \text{ heads of size } d_h = 128\\
K,\ V &= W_K\, u, \quad W_V\, u && H_{kv} = 8 \text{ key/value heads}\\
Q,\ K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position}\\
S_h &= s\, Q_h K_{g(h)}^\top + M && \text{mask } M\text{: causal}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\, V_{g(h)} &&\\
\mathrm{Attn}(u) &= W_O\, [O_1; \ldots; O_H] &&\\
\mathrm{FFN}(u) &= W_{\text{down}}\big(\,\mathrm{SiLU}(W_{\text{gate}}\, u) \odot W_{\text{up}}\, u\big) && \text{gated MLP, width 14336}
\end{aligned}$$

Parameters of one block ($|\theta_\ell| = 218{,}112{,}000$): $W_Q \in \mathbb{R}^{4096\times4096}$, $W_K \in \mathbb{R}^{1024\times4096}$, $W_V \in \mathbb{R}^{1024\times4096}$, $W_O \in \mathbb{R}^{4096\times4096}$, $W_{\text{gate}} \in \mathbb{R}^{14336\times4096}$, $W_{\text{up}} \in \mathbb{R}^{14336\times4096}$, $W_{\text{down}} \in \mathbb{R}^{4096\times14336}$, plus 8,192 parameters in norm scales, biases and other vectors. This type is used 32 times.

Definition 2.6 (Output head)**.** $\Psi : \mathbb{R}^{T\times d} \to \mathbb{R}^{T\times V}$ normalises the last hidden state and scores every vocabulary token with its own matrix $W_{\mathrm{LM}} \in \mathbb{R}^{V\times d}$:

$$\Psi(h)_t = W_{\mathrm{LM}}\ \mathrm{RMSNorm}_{\text{final}}(h)_t \tag{6}$$

2.3 Neurons and attention heads

Definition 2.7 (Neuron, pre-activation value and state)**.** The feed-forward sub-layer of block ℓ contains $d_{ff}^{(\ell)} = 14336$ neurons. Neuron j of layer ℓ is the scalar function of a normalised token vector $u \in \mathbb{R}^d$

$$\eta_j^{(\ell)}(u) = \mathrm{SiLU}\left(W_{\mathrm{gate},j}^{(\ell)} \cdot u\right)\left(W_{\mathrm{up},j}^{(\ell)} \cdot u\right), \tag{7}$$

where $W_{\mathrm{gate},j}$ is the j-th row of W_{gate}. The feed-forward output is the sum of the neuron outputs, each written along its own column of W_{down}:

$$\mathrm{FFN}_\ell(u) = \sum_{j=1}^{d_{ff}^{(\ell)}} \eta_j^{(\ell)}(u)\, W_{\mathrm{down}}^{(\ell)}[:, j]. \tag{8}$$

Across its L blocks the network has $N = \sum_{\ell=1}^{L} d_{ff}^{(\ell)} = 458{,}752$ feed-forward neurons. For input x at position t, with $u_t^{(\ell)}(x)$ the normalised input of the feed-forward sub-layer:

- $\mathcal{V}(\eta; x, t) = W_{\mathrm{gate},j}^{(\ell)} \cdot u_t^{(\ell)}(x) \in \mathbb{R}$ is the **pre-activation value** of neuron η (the argument of the non-linearity).
- $\mathcal{S}(\eta; x, t)$ is its **activation state**: *active* if $\mathcal{V} > 0$ and *inactive* if $\mathcal{V} \le 0$. The non-linearity is smooth, unlike ReLU, so an inactive neuron's output is strongly suppressed rather than exactly zero.

Definition 2.8 (Attention head and attention pattern)**.** Head h of an attention layer ℓ reads from the whole sequence through its attention pattern $P_h^{(\ell)}(x) \in \mathbb{R}^{T \times T}$ (row t = how much position t attends to each earlier position) and writes through its own $d \times d_h$ slice $W_{O,h}^{(\ell)}$ of W_O:

$$\mathrm{Attn}_\ell(u) = \sum_{h=1}^{H} P_h^{(\ell)}\, V_{g(h)}\, W_{O,h}^{(\ell)\top}. \tag{9}$$

$g(h)$ is the key/value head shared by query head h (grouped-query attention). The pattern P_h plays the role that the activation state plays for neurons: once P_h is fixed, the head is linear in its values. The network has 32 attention layers $\times$ $H = 32$ heads $= 1024$ attention heads.

2.4 Architecture and parameters

Definition 2.9 (Architecture and parameters)**.** The **architecture** is the tuple of sizes and the layer-type sequence:

$$\mathcal{A} = (V, d, L, c, H, H_{kv}, d_h, d_{ff}) = (128{,}256,\ 4096,\ 32,\ (\mathrm{D}^{32}),\ 32,\ 8,\ 128,\ 14336). \tag{10}$$

The **parameters** are $\theta = \{W_E,\ \theta_1, \ldots, \theta_L,\ \gamma_{\mathrm{final}},\ W_{\mathrm{LM}}\}$. Grouping the blocks by type,

$$\begin{aligned} |\theta| &= \sum_{\text{types } c} n_c\, |\theta_c| + |\theta_{\mathrm{rest}}| \\ &= 32 \times 218{,}112{,}000 + 1{,}050{,}677{,}248 \\ &= 8{,}030{,}261{,}248, \end{aligned}$$

where n_c is how often type c occurs and θ_{rest} holds everything outside the repeated stacks (embedding, final norm, ...; Table 3).

2.5 Output representations

Definition 2.10 (Output representations)**.** For an input sequence x, the output can be read at three levels:

1. **Logits:** $y = f_\theta(x) \in \mathbb{R}^{T \times V}$, one score per vocabulary token at every position.
2. **Next-token distribution:** $p(x)_{t,k} = \dfrac{\exp(y_{t,k})}{\sum_{j=1}^{V} \exp(y_{t,j})} = \Pr\left(x_{t+1} = k \mid x_1, \ldots, x_t\right)$.
3. **Predicted next token:** $z(x) = \mathrm{argmax}_{k \in \{1,\ldots,V\}}\, y_{T,k}$ (greedy decoding). Generation appends $z(x)$ (or a sample from $p(x)_T$) to x and applies f_θ again; in practice the keys and values of earlier positions are cached.

The model is **causal**: y_t depends only on $x_1, \ldots, x_t$, because every attention mask hides later positions. Training minimises the cross-entropy $-\sum_t \log p(x)_{t, x_{t+1}}$.

2.6 Intermediate states and sub-network decomposition

To study what happens at one layer $\ell \in \{1, \dots, L\}$, split the composition into a prefix, the layer itself and a suffix:

$$f_\theta = \underbrace{\Psi \circ \mathcal{B}^{(L)} \circ \cdots \circ \mathcal{B}^{(\ell+1)}}_{G_{\ell+1}\ \text{(suffix)}} \circ \underbrace{\mathcal{B}^{(\ell)}}_{\text{layer } \ell} \circ \underbrace{\mathcal{B}^{(\ell-1)} \circ \cdots \circ \mathcal{B}^{(1)} \circ E}_{F_{\ell-1}\ \text{(prefix)}} \tag{11}$$

- **Prefix** $F_{\ell-1} : \mathcal{X} \to \mathbb{R}^{T \times d}$ maps the tokens to the hidden state entering layer ℓ ($F_0 = E$).
- **Hidden state** (residual stream) after layer ℓ: $h^{(\ell)} = F_\ell(x) = \mathcal{B}^{(\ell)}(h^{(\ell-1)}) \in \mathbb{R}^{T \times d}$, with $h^{(0)} = E(x)$.
- **Inside layer ℓ:** the attention write $a^{(\ell)} = \mathrm{Attn}_\ell\big(\mathrm{RMSNorm}_1(h^{(\ell-1)})\big)$, the mid-block state $\hat{h}^{(\ell)} = h^{(\ell-1)} + a^{(\ell)}$, the feed-forward input $u^{(\ell)} = \mathrm{RMSNorm}_2(\hat{h}^{(\ell)})$, the pre-activations $\hat{x}^{(\ell)} = u^{(\ell)} W_{\text{gate}}^{(\ell)\top} \in \mathbb{R}^{T \times d_{ff}}$ and the neuron outputs $\eta^{(\ell)}(u^{(\ell)}) \in \mathbb{R}^{T \times d_{ff}}$, and the feed-forward write $m^{(\ell)} = \mathrm{FFN}_\ell(u^{(\ell)})$.
- **Suffix** $G_{\ell+1} : \mathbb{R}^{T \times d} \to \mathbb{R}^{T \times V}$ maps $h^{(\ell)}$ to the logits ($G_{L+1} = \Psi$).

Residual stream. Because every sub-layer only *adds* to the hidden state, unrolling the composition gives the final hidden state as the embedding plus everything the layers wrote:

$$h^{(L)} = h^{(0)} + \sum_{\ell=1}^{L} \big(a^{(\ell)} + m^{(\ell)}\big) \tag{12}$$

So $f_\theta(x) = \Psi(h^{(L)})$: each layer reads the current stream and adds a correction to it.

2.7 Local linear representation

A ReLU network is piecewise affine: inside a polytope cell it collapses to one matrix. This model is not, because softmax, SiLU and the normalisations are smooth. Its local behaviour is described by the first-order (Jacobian) linearisation instead. Tokens are discrete, so we perturb the embedded input $h^{(0)}$.

Definition 2.11 (Local linear response)**.** Let $J_\ell = D\mathcal{B}^{(\ell)}(h^{(\ell-1)})$ be the Jacobian of block ℓ at the current hidden state (tensors flattened to vectors). The chain rule applied to the residual sub-layers $(\mathrm{id} + g_{\ell,i})$ gives

$$J_\ell = (I + Dg_{\ell,2})(I + Dg_{\ell,1}), \tag{13}$$

where $g_{\ell,1}$ is the attention branch, $g_{\ell,2}$ the feed-forward branch and each Dg is taken at that branch's input. For a small perturbation δ of $h^{(0)}$, the response of the network decomposed around layer ℓ is

$$f_\theta(h^{(0)} + \delta) - f_\theta(h^{(0)}) \approx G_x^{(\ell+1)} \cdot J_\ell \cdot F_x^{(\ell-1)} \cdot \delta, \tag{14}$$

with $F_x^{(\ell-1)} = J_{\ell-1} \cdots J_1$ (prefix) and $G_x^{(\ell+1)} = J_\Psi\, J_L \cdots J_{\ell+1}$ (suffix). Multiplying out the products of $(I + Dg)$ terms writes $J_\Psi J_L \cdots J_1$ as a sum over *paths*: the identity path straight from embedding to head, plus one term for every subset of sub-layers a signal passes through.

2.8 Summary of notation

Symbol	Meaning
$f_\theta : \mathcal{X} \to \mathcal{Y}$	the whole model, $\{1..V\}^T \to \mathbb{R}^{T \times V}$
$E,\ \Psi$	embedding and output head
$\mathcal{B}^{(\ell)},\ \theta_\ell$	block ℓ and its parameters, $\ell = 1, \dots, L$
$\mathcal{B}_c,\ c(\ell)$	block type c; type of layer ℓ
$\mathcal{B}^{\circ n}$	n consecutive blocks of one type (each with its own weights)
$(\mathrm{id} + g)$	residual sub-layer $h \mapsto h + g(h)$
$h^{(\ell)}$	hidden state (residual stream) after layer ℓ; $h^{(0)}$ = embeddings
$a^{(\ell)},\ m^{(\ell)}$	what the attention / feed-forward sub-layer of layer ℓ adds to the stream
$\eta_j^{(\ell)}$	j-th feed-forward neuron of layer ℓ
$\mathcal{V}(\eta; x, t),\ \mathcal{S}(\eta; x, t)$	pre-activation value and active/inactive state of a neuron
$P_h^{(\ell)}$	attention pattern of head h in layer ℓ ($T \times T$)

continued on next page

Symbol	Meaning
$F_{\ell-1},\ G_{\ell+1}$	prefix (embedding to layer $\ell-1$) and suffix (layer $\ell+1$ to logits)
$J_\ell,\ F_x^{(\ell-1)},\ G_x^{(\ell+1)}$	Jacobian of block ℓ, and of the prefix / suffix at input x
$p(x),\ z(x)$	next-token distribution and greedy next-token prediction

3 Forward Pass in Pseudocode

Algorithm 1 Llama-3.1-8B-Instruct: full forward pass

Require: token ids x of shape $B \times T$

$h \leftarrow E(x)$ ▷ $h^{(0)}$: one vector per token, $B{\times}T{\times}4096$

for $\ell \leftarrow 1$ **to** 32 **do** ▷ 32 blocks of type $\mathcal{B}_{\mathrm{D}}$

  $h \leftarrow \mathcal{B}_{\mathrm{D}}(h;\, \theta_\ell)$ ▷ Decoder block; result is $h^{(\ell)}$

end for

$h \leftarrow \mathrm{RMSNorm}_{\mathrm{final}}(h)$ ▷ $h = h^{(L)}$; this and the next step are Ψ

$y \leftarrow h\, W_{\mathrm{LM}}^{\top}$ ▷ logits $B{\times}T{\times}128,256$

return y ▷ softmax(y) = next-token probabilities

Algorithm 2 Block $\mathcal{B}_{\mathrm{D}}(h)$ — Decoder block

Require: h of shape $B \times T \times 4096$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection

$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector

token mixing: multi-head self-attention

$Q \leftarrow uW_Q^{\top}$ ▷ $B{\times}T{\times}32{\times}128$ (32 query heads)

$K \leftarrow uW_K^{\top};\ V \leftarrow uW_V^{\top}$ ▷ $B{\times}T{\times}8{\times}128$ (8 key/value heads, each shared by 4 query heads)

$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs

$P \leftarrow \mathrm{softmax}(s\, QK^{\top} + M)$ ▷ mask: causal: a token sees only earlier tokens

$O \leftarrow PV$ ▷ $B{\times}T{\times}32{\times}128$

$m \leftarrow O\, W_O^{\top}$ ▷ concatenate heads, project back to $B{\times}T{\times}4096$

$h \leftarrow r + m$ ▷ add residual

channel mixing: feed-forward network (each token independently)

$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$

$m \leftarrow \big(\mathrm{SiLU}(uW_{\mathrm{gate}}^{\top}) \odot uW_{\mathrm{up}}^{\top}\big)W_{\mathrm{down}}^{\top}$ ▷ expand to 14336, gate, shrink back to 4096

$h \leftarrow r + m$ ▷ add residual

return h

4 Compact Architecture (repeated layers written once)

4.1 Modules outside the repeated stacks

Table 3: Weights that appear once

Weight	Type	Shape	Params
`model.embed_tokens.weight`	Embedding	128256×4096	525,336,576
`model.norm.weight`	LlamaRMSNorm	4096	4,096
`lm_head.weight`	Linear	128256×4096	525,336,576

4.2 Language-model decoder: `model.layers`

Table 4: $\mathcal{B}_{\mathrm{D}}$: Decoder block — layers 0–31; weight names are relative to `model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	4096×4096	16,777,216

continued on next page

Weight (relative)	Type	Shape	Params
`self_attn.k_proj.weight`	Linear	1024×4096	4,194,304
`self_attn.v_proj.weight`	Linear	1024×4096	4,194,304
`self_attn.o_proj.weight`	Linear	4096×4096	16,777,216
`mlp.gate_proj.weight`	Linear	14336×4096	58,720,256
`mlp.up_proj.weight`	Linear	14336×4096	58,720,256
`mlp.down_proj.weight`	Linear	4096×14336	58,720,256
`input_layernorm.weight`	LlamaRMSNorm	4096	4,096
`post_attention_layernorm.weight`	LlamaRMSNorm	4096	4,096
Per block			**218,112,000**
× 32 blocks			**6,979,584,000**

5 What Each Layer Does

One row per *kind* of layer present in this model. "Appears as" gives the module names (layer indices removed).

Layer (appears as)	What it does
Language model	
Token embedding `embed_tokens`	Lookup table with one learned vector per vocabulary token. Turns token ids into the vectors the network computes with (Gemma multiplies them by $\sqrt{d}$).
Query projection (q_proj) `self_attn.q_proj`	Turns each token vector into query vectors, one per head: "what am I looking for?".
Key projection (k_proj) `self_attn.k_proj`	Turns each token into key vectors: "what do I contain?". Queries are compared with keys to decide where to attend. With grouped-query attention there are fewer key heads than query heads.
Value projection (v_proj) `self_attn.v_proj`	Turns each token into value vectors: the information actually copied to tokens that attend to it.
Output projection (o_proj) `self_attn.o_proj`	Concatenates all heads' results and mixes them back into one hidden-size vector per token.
LM head (output projection) `lm_head`	Maps each final hidden vector to one score (logit) per vocabulary token. Softmax of the logits gives the next-token probabilities. When tied, it reuses the embedding matrix.
Gate projection (gate_proj) `mlp.gate_proj`	Expands the token vector to the MLP width and passes it through SiLU/GELU. The result acts as a soft on/off switch for each hidden unit.
Up projection (up_proj) `mlp.up_proj`	Expands the token vector to the MLP width (the "content" path). It is multiplied element-wise by the gate (SwiGLU/GeGLU).
Down projection (down_proj) `mlp.down_proj`	Shrinks the gated hidden vector back to the model width so it can be added to the residual stream.
Input norm (input_layernorm) `layers.input_layernorm`	RMSNorm applied to the residual stream before attention. Rescales each token vector to unit RMS, then multiplies by a learned per-channel weight.
post_attention_layernorm `layers.post_attention_layernorm`	Despite the name, this is the pre-MLP norm: RMSNorm on the residual stream after attention, feeding the MLP.
Final norm `norm`	RMSNorm applied once after the last decoder layer, just before the LM head.

6 Parameter Count

Table 6: Parameters by component

Component	Parameters	% of total
MLP / MoE	5,637,144,576	70.2%
Attention / token mixer	1,342,177,280	16.7%
Token embedding	525,336,576	6.5%

continued on next page

Component	Parameters	% of total
LM head	525,336,576	6.5%
Norms	266,240	0.0%
Total	**8,030,261,248**	**100.0%**

Table 7: Parameters by block type

Block type	Count	Params / block	Total
$\mathcal{B}_{\mathrm{D}}$ Decoder block	32	218,112,000	6,979,584,000
Modules outside stacks	1	–	1,050,677,248
Total			**8,030,261,248**

Appendix A Full Layer-by-Layer Table

Every weight tensor with every repetition written out (the original table).

Layer path	Type	Shape	Dtype	Params
`lm_head.weight`	Linear	128256 × 4096	bfloat16	525,336,576
`model`	LlamaModel	–	–	–
`model.embed_tokens.weight`	Embedding	128256 × 4096	bfloat16	525,336,576
`model.layers.0`	LlamaDecoderLayer	–	–	–
`model.layers.0.input_layernorm.weight`	LlamaRMSNorm	4096	bfloat16	4,096
`model.layers.0.mlp`	LlamaMLP	–	–	–
`model.layers.0.mlp.down_proj.weight`	Linear	4096 × 14336	bfloat16	58,720,256
`model.layers.0.mlp.gate_proj.weight`	Linear	14336 × 4096	bfloat16	58,720,256
`model.layers.0.mlp.up_proj.weight`	Linear	14336 × 4096	bfloat16	58,720,256
`model.layers.0.post_attention_layernorm.weight`	LlamaRMSNorm	4096	bfloat16	4,096
`model.layers.0.self_attn`	LlamaAttention	–	–	–
`model.layers.0.self_attn.k_proj.weight`	Linear	1024 × 4096	bfloat16	4,194,304
`model.layers.0.self_attn.o_proj.weight`	Linear	4096 × 4096	bfloat16	16,777,216
`model.layers.0.self_attn.q_proj.weight`	Linear	4096 × 4096	bfloat16	16,777,216
`model.layers.0.self_attn.v_proj.weight`	Linear	1024 × 4096	bfloat16	4,194,304
`model.layers.1`	LlamaDecoderLayer	–	–	–
`model.layers.1.input_layernorm.weight`	LlamaRMSNorm	4096	bfloat16	4,096
`model.layers.1.mlp`	LlamaMLP	–	–	–
`model.layers.1.mlp.down_proj.weight`	Linear	4096 × 14336	bfloat16	58,720,256
`model.layers.1.mlp.gate_proj.weight`	Linear	14336 × 4096	bfloat16	58,720,256
`model.layers.1.mlp.up_proj.weight`	Linear	14336 × 4096	bfloat16	58,720,256
`model.layers.1.post_attention_layernorm.weight`	LlamaRMSNorm	4096	bfloat16	4,096
`model.layers.1.self_attn`	LlamaAttention	–	–	–
`model.layers.1.self_attn.k_proj.weight`	Linear	1024 × 4096	bfloat16	4,194,304
`model.layers.1.self_attn.o_proj.weight`	Linear	4096 × 4096	bfloat16	16,777,216
`model.layers.1.self_attn.q_proj.weight`	Linear	4096 × 4096	bfloat16	16,777,216
`model.layers.1.self_attn.v_proj.weight`	Linear	1024 × 4096	bfloat16	4,194,304
`model.layers.2`	LlamaDecoderLayer	–	–	–
`model.layers.2.input_layernorm.weight`	LlamaRMSNorm	4096	bfloat16	4,096
`model.layers.2.mlp`	LlamaMLP	–	–	–
`model.layers.2.mlp.down_proj.weight`	Linear	4096 × 14336	bfloat16	58,720,256
`model.layers.2.mlp.gate_proj.weight`	Linear	14336 × 4096	bfloat16	58,720,256
`model.layers.2.mlp.up_proj.weight`	Linear	14336 × 4096	bfloat16	58,720,256
`model.layers.2.post_attention_layernorm.weight`	LlamaRMSNorm	4096	bfloat16	4,096
`model.layers.2.self_attn`	LlamaAttention	–	–	–
`model.layers.2.self_attn.k_proj.weight`	Linear	1024 × 4096	bfloat16	4,194,304
`model.layers.2.self_attn.o_proj.weight`	Linear	4096 × 4096	bfloat16	16,777,216
`model.layers.2.self_attn.q_proj.weight`	Linear	4096 × 4096	bfloat16	16,777,216
`model.layers.2.self_attn.v_proj.weight`	Linear	1024 × 4096	bfloat16	4,194,304
`model.layers.3`	LlamaDecoderLayer	–	–	–
`model.layers.3.input_layernorm.weight`	LlamaRMSNorm	4096	bfloat16	4,096
`model.layers.3.mlp`	LlamaMLP	–	–	–
`model.layers.3.mlp.down_proj.weight`	Linear	4096 × 14336	bfloat16	58,720,256
`model.layers.3.mlp.gate_proj.weight`	Linear	14336 × 4096	bfloat16	58,720,256
`model.layers.3.mlp.up_proj.weight`	Linear	14336 × 4096	bfloat16	58,720,256
`model.layers.3.post_attention_layernorm.weight`	LlamaRMSNorm	4096	bfloat16	4,096
`model.layers.3.self_attn`	LlamaAttention	–	–	–

continued on next page

Layer path	Type	Shape	Dtype	Params
model.layers.3.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.3.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.3.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.3.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.4	LlamaDecoderLayer	–	–	–
model.layers.4.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.4.mlp	LlamaMLP	–	–	–
model.layers.4.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.4.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.4.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.4.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.4.self_attn	LlamaAttention	–	–	–
model.layers.4.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.4.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.4.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.4.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.5	LlamaDecoderLayer	–	–	–
model.layers.5.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.5.mlp	LlamaMLP	–	–	–
model.layers.5.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.5.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.5.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.5.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.5.self_attn	LlamaAttention	–	–	–
model.layers.5.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.5.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.5.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.5.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.6	LlamaDecoderLayer	–	–	–
model.layers.6.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.6.mlp	LlamaMLP	–	–	–
model.layers.6.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.6.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.6.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.6.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.6.self_attn	LlamaAttention	–	–	–
model.layers.6.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.6.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.6.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.6.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.7	LlamaDecoderLayer	–	–	–
model.layers.7.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.7.mlp	LlamaMLP	–	–	–
model.layers.7.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.7.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.7.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.7.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.7.self_attn	LlamaAttention	–	–	–
model.layers.7.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.7.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.7.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.7.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.8	LlamaDecoderLayer	–	–	–
model.layers.8.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.8.mlp	LlamaMLP	–	–	–
model.layers.8.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.8.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.8.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.8.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.8.self_attn	LlamaAttention	–	–	–
model.layers.8.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.8.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.8.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.8.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.9	LlamaDecoderLayer	–	–	–
model.layers.9.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.9.mlp	LlamaMLP	–	–	–
model.layers.9.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.9.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256

continued on next page

Layer path	Type	Shape	Dtype	Params
model.layers.9.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.9.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.9.self_attn	LlamaAttention	–	–	–
model.layers.9.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.9.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.9.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.9.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.10	LlamaDecoderLayer	–	–	–
model.layers.10.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.10.mlp	LlamaMLP	–	–	–
model.layers.10.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.10.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.10.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.10.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.10.self_attn	LlamaAttention	–	–	–
model.layers.10.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.10.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.10.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.10.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.11	LlamaDecoderLayer	–	–	–
model.layers.11.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.11.mlp	LlamaMLP	–	–	–
model.layers.11.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.11.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.11.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.11.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.11.self_attn	LlamaAttention	–	–	–
model.layers.11.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.11.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.11.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.11.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.12	LlamaDecoderLayer	–	–	–
model.layers.12.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.12.mlp	LlamaMLP	–	–	–
model.layers.12.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.12.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.12.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.12.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.12.self_attn	LlamaAttention	–	–	–
model.layers.12.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.12.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.12.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.12.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.13	LlamaDecoderLayer	–	–	–
model.layers.13.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.13.mlp	LlamaMLP	–	–	–
model.layers.13.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.13.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.13.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.13.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.13.self_attn	LlamaAttention	–	–	–
model.layers.13.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.13.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.13.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.13.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.14	LlamaDecoderLayer	–	–	–
model.layers.14.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.14.mlp	LlamaMLP	–	–	–
model.layers.14.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.14.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.14.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.14.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.14.self_attn	LlamaAttention	–	–	–
model.layers.14.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.14.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.14.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.14.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.15	LlamaDecoderLayer	–	–	–

continued on next page

Layer path	Type	Shape	Dtype	Params
model.layers.15.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.15.mlp	LlamaMLP	–	–	–
model.layers.15.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.15.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.15.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.15.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.15.self_attn	LlamaAttention	–	–	–
model.layers.15.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.15.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.15.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.15.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.16	LlamaDecoderLayer	–	–	–
model.layers.16.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.16.mlp	LlamaMLP	–	–	–
model.layers.16.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.16.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.16.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.16.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.16.self_attn	LlamaAttention	–	–	–
model.layers.16.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.16.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.16.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.16.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.17	LlamaDecoderLayer	–	–	–
model.layers.17.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.17.mlp	LlamaMLP	–	–	–
model.layers.17.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.17.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.17.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.17.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.17.self_attn	LlamaAttention	–	–	–
model.layers.17.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.17.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.17.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.17.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.18	LlamaDecoderLayer	–	–	–
model.layers.18.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.18.mlp	LlamaMLP	–	–	–
model.layers.18.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.18.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.18.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.18.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.18.self_attn	LlamaAttention	–	–	–
model.layers.18.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.18.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.18.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.18.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.19	LlamaDecoderLayer	–	–	–
model.layers.19.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.19.mlp	LlamaMLP	–	–	–
model.layers.19.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.19.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.19.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.19.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.19.self_attn	LlamaAttention	–	–	–
model.layers.19.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.19.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.19.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.19.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.20	LlamaDecoderLayer	–	–	–
model.layers.20.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.20.mlp	LlamaMLP	–	–	–
model.layers.20.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.20.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.20.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.20.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.20.self_attn	LlamaAttention	–	–	–
model.layers.20.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304

continued on next page

Layer path	Type	Shape	Dtype	Params
model.layers.20.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.20.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.20.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.21	LlamaDecoderLayer	–	–	–
model.layers.21.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.21.mlp	LlamaMLP	–	–	–
model.layers.21.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.21.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.21.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.21.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.21.self_attn	LlamaAttention	–	–	–
model.layers.21.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.21.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.21.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.21.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.22	LlamaDecoderLayer	–	–	–
model.layers.22.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.22.mlp	LlamaMLP	–	–	–
model.layers.22.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.22.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.22.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.22.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.22.self_attn	LlamaAttention	–	–	–
model.layers.22.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.22.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.22.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.22.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.23	LlamaDecoderLayer	–	–	–
model.layers.23.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.23.mlp	LlamaMLP	–	–	–
model.layers.23.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.23.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.23.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.23.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.23.self_attn	LlamaAttention	–	–	–
model.layers.23.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.23.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.23.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.23.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.24	LlamaDecoderLayer	–	–	–
model.layers.24.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.24.mlp	LlamaMLP	–	–	–
model.layers.24.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.24.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.24.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.24.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.24.self_attn	LlamaAttention	–	–	–
model.layers.24.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.24.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.24.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.24.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.25	LlamaDecoderLayer	–	–	–
model.layers.25.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.25.mlp	LlamaMLP	–	–	–
model.layers.25.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.25.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.25.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.25.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.25.self_attn	LlamaAttention	–	–	–
model.layers.25.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.25.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.25.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.25.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.26	LlamaDecoderLayer	–	–	–
model.layers.26.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.26.mlp	LlamaMLP	–	–	–
model.layers.26.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.26.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.26.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256

continued on next page

Layer path	Type	Shape	Dtype	Params
model.layers.26.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.26.self_attn	LlamaAttention	–	–	–
model.layers.26.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.26.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.26.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.26.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.27	LlamaDecoderLayer	–	–	–
model.layers.27.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.27.mlp	LlamaMLP	–	–	–
model.layers.27.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.27.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.27.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.27.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.27.self_attn	LlamaAttention	–	–	–
model.layers.27.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.27.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.27.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.27.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.28	LlamaDecoderLayer	–	–	–
model.layers.28.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.28.mlp	LlamaMLP	–	–	–
model.layers.28.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.28.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.28.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.28.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.28.self_attn	LlamaAttention	–	–	–
model.layers.28.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.28.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.28.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.28.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.29	LlamaDecoderLayer	–	–	–
model.layers.29.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.29.mlp	LlamaMLP	–	–	–
model.layers.29.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.29.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.29.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.29.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.29.self_attn	LlamaAttention	–	–	–
model.layers.29.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.29.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.29.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.29.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.30	LlamaDecoderLayer	–	–	–
model.layers.30.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.30.mlp	LlamaMLP	–	–	–
model.layers.30.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.30.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.30.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.30.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.30.self_attn	LlamaAttention	–	–	–
model.layers.30.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.30.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.30.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.30.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.31	LlamaDecoderLayer	–	–	–
model.layers.31.input_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.31.mlp	LlamaMLP	–	–	–
model.layers.31.mlp.down_proj.weight	Linear	4096 × 14336	bfloat16	58,720,256
model.layers.31.mlp.gate_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.31.mlp.up_proj.weight	Linear	14336 × 4096	bfloat16	58,720,256
model.layers.31.post_attention_layernorm.weight	LlamaRMSNorm	4096	bfloat16	4,096
model.layers.31.self_attn	LlamaAttention	–	–	–
model.layers.31.self_attn.k_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.layers.31.self_attn.o_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.31.self_attn.q_proj.weight	Linear	4096 × 4096	bfloat16	16,777,216
model.layers.31.self_attn.v_proj.weight	Linear	1024 × 4096	bfloat16	4,194,304
model.norm.weight	LlamaRMSNorm	4096	bfloat16	4,096

Appendix B Per-Layer Parameter Count

Module	Params	Running total
lm_head	525,336,576	525,336,576
model.embed_tokens	525,336,576	1,050,673,152
model.layers.0.input_layernorm	4,096	1,050,677,248
model.layers.0.mlp.down_proj	58,720,256	1,109,397,504
model.layers.0.mlp.gate_proj	58,720,256	1,168,117,760
model.layers.0.mlp.up_proj	58,720,256	1,226,838,016
model.layers.0.post_attention_layernorm	4,096	1,226,842,112
model.layers.0.self_attn.k_proj	4,194,304	1,231,036,416
model.layers.0.self_attn.o_proj	16,777,216	1,247,813,632
model.layers.0.self_attn.q_proj	16,777,216	1,264,590,848
model.layers.0.self_attn.v_proj	4,194,304	1,268,785,152
model.layers.1.input_layernorm	4,096	1,268,789,248
model.layers.1.mlp.down_proj	58,720,256	1,327,509,504
model.layers.1.mlp.gate_proj	58,720,256	1,386,229,760
model.layers.1.mlp.up_proj	58,720,256	1,444,950,016
model.layers.1.post_attention_layernorm	4,096	1,444,954,112
model.layers.1.self_attn.k_proj	4,194,304	1,449,148,416
model.layers.1.self_attn.o_proj	16,777,216	1,465,925,632
model.layers.1.self_attn.q_proj	16,777,216	1,482,702,848
model.layers.1.self_attn.v_proj	4,194,304	1,486,897,152
model.layers.2.input_layernorm	4,096	1,486,901,248
model.layers.2.mlp.down_proj	58,720,256	1,545,621,504
model.layers.2.mlp.gate_proj	58,720,256	1,604,341,760
model.layers.2.mlp.up_proj	58,720,256	1,663,062,016
model.layers.2.post_attention_layernorm	4,096	1,663,066,112
model.layers.2.self_attn.k_proj	4,194,304	1,667,260,416
model.layers.2.self_attn.o_proj	16,777,216	1,684,037,632
model.layers.2.self_attn.q_proj	16,777,216	1,700,814,848
model.layers.2.self_attn.v_proj	4,194,304	1,705,009,152
model.layers.3.input_layernorm	4,096	1,705,013,248
model.layers.3.mlp.down_proj	58,720,256	1,763,733,504
model.layers.3.mlp.gate_proj	58,720,256	1,822,453,760
model.layers.3.mlp.up_proj	58,720,256	1,881,174,016
model.layers.3.post_attention_layernorm	4,096	1,881,178,112
model.layers.3.self_attn.k_proj	4,194,304	1,885,372,416
model.layers.3.self_attn.o_proj	16,777,216	1,902,149,632
model.layers.3.self_attn.q_proj	16,777,216	1,918,926,848
model.layers.3.self_attn.v_proj	4,194,304	1,923,121,152
model.layers.4.input_layernorm	4,096	1,923,125,248
model.layers.4.mlp.down_proj	58,720,256	1,981,845,504
model.layers.4.mlp.gate_proj	58,720,256	2,040,565,760
model.layers.4.mlp.up_proj	58,720,256	2,099,286,016
model.layers.4.post_attention_layernorm	4,096	2,099,290,112
model.layers.4.self_attn.k_proj	4,194,304	2,103,484,416
model.layers.4.self_attn.o_proj	16,777,216	2,120,261,632
model.layers.4.self_attn.q_proj	16,777,216	2,137,038,848
model.layers.4.self_attn.v_proj	4,194,304	2,141,233,152
model.layers.5.input_layernorm	4,096	2,141,237,248
model.layers.5.mlp.down_proj	58,720,256	2,199,957,504
model.layers.5.mlp.gate_proj	58,720,256	2,258,677,760
model.layers.5.mlp.up_proj	58,720,256	2,317,398,016
model.layers.5.post_attention_layernorm	4,096	2,317,402,112
model.layers.5.self_attn.k_proj	4,194,304	2,321,596,416
model.layers.5.self_attn.o_proj	16,777,216	2,338,373,632
model.layers.5.self_attn.q_proj	16,777,216	2,355,150,848
model.layers.5.self_attn.v_proj	4,194,304	2,359,345,152

continued on next page

Module	Params	Running total
model.layers.6.input_layernorm	4,096	2,359,349,248
model.layers.6.mlp.down_proj	58,720,256	2,418,069,504
model.layers.6.mlp.gate_proj	58,720,256	2,476,789,760
model.layers.6.mlp.up_proj	58,720,256	2,535,510,016
model.layers.6.post_attention_layernorm	4,096	2,535,514,112
model.layers.6.self_attn.k_proj	4,194,304	2,539,708,416
model.layers.6.self_attn.o_proj	16,777,216	2,556,485,632
model.layers.6.self_attn.q_proj	16,777,216	2,573,262,848
model.layers.6.self_attn.v_proj	4,194,304	2,577,457,152
model.layers.7.input_layernorm	4,096	2,577,461,248
model.layers.7.mlp.down_proj	58,720,256	2,636,181,504
model.layers.7.mlp.gate_proj	58,720,256	2,694,901,760
model.layers.7.mlp.up_proj	58,720,256	2,753,622,016
model.layers.7.post_attention_layernorm	4,096	2,753,626,112
model.layers.7.self_attn.k_proj	4,194,304	2,757,820,416
model.layers.7.self_attn.o_proj	16,777,216	2,774,597,632
model.layers.7.self_attn.q_proj	16,777,216	2,791,374,848
model.layers.7.self_attn.v_proj	4,194,304	2,795,569,152
model.layers.8.input_layernorm	4,096	2,795,573,248
model.layers.8.mlp.down_proj	58,720,256	2,854,293,504
model.layers.8.mlp.gate_proj	58,720,256	2,913,013,760
model.layers.8.mlp.up_proj	58,720,256	2,971,734,016
model.layers.8.post_attention_layernorm	4,096	2,971,738,112
model.layers.8.self_attn.k_proj	4,194,304	2,975,932,416
model.layers.8.self_attn.o_proj	16,777,216	2,992,709,632
model.layers.8.self_attn.q_proj	16,777,216	3,009,486,848
model.layers.8.self_attn.v_proj	4,194,304	3,013,681,152
model.layers.9.input_layernorm	4,096	3,013,685,248
model.layers.9.mlp.down_proj	58,720,256	3,072,405,504
model.layers.9.mlp.gate_proj	58,720,256	3,131,125,760
model.layers.9.mlp.up_proj	58,720,256	3,189,846,016
model.layers.9.post_attention_layernorm	4,096	3,189,850,112
model.layers.9.self_attn.k_proj	4,194,304	3,194,044,416
model.layers.9.self_attn.o_proj	16,777,216	3,210,821,632
model.layers.9.self_attn.q_proj	16,777,216	3,227,598,848
model.layers.9.self_attn.v_proj	4,194,304	3,231,793,152
model.layers.10.input_layernorm	4,096	3,231,797,248
model.layers.10.mlp.down_proj	58,720,256	3,290,517,504
model.layers.10.mlp.gate_proj	58,720,256	3,349,237,760
model.layers.10.mlp.up_proj	58,720,256	3,407,958,016
model.layers.10.post_attention_layernorm	4,096	3,407,962,112
model.layers.10.self_attn.k_proj	4,194,304	3,412,156,416
model.layers.10.self_attn.o_proj	16,777,216	3,428,933,632
model.layers.10.self_attn.q_proj	16,777,216	3,445,710,848
model.layers.10.self_attn.v_proj	4,194,304	3,449,905,152
model.layers.11.input_layernorm	4,096	3,449,909,248
model.layers.11.mlp.down_proj	58,720,256	3,508,629,504
model.layers.11.mlp.gate_proj	58,720,256	3,567,349,760
model.layers.11.mlp.up_proj	58,720,256	3,626,070,016
model.layers.11.post_attention_layernorm	4,096	3,626,074,112
model.layers.11.self_attn.k_proj	4,194,304	3,630,268,416
model.layers.11.self_attn.o_proj	16,777,216	3,647,045,632
model.layers.11.self_attn.q_proj	16,777,216	3,663,822,848
model.layers.11.self_attn.v_proj	4,194,304	3,668,017,152
model.layers.12.input_layernorm	4,096	3,668,021,248
model.layers.12.mlp.down_proj	58,720,256	3,726,741,504
model.layers.12.mlp.gate_proj	58,720,256	3,785,461,760
model.layers.12.mlp.up_proj	58,720,256	3,844,182,016

continued on next page

Module	Params	Running total
model.layers.12.post_attention_layernorm	4,096	3,844,186,112
model.layers.12.self_attn.k_proj	4,194,304	3,848,380,416
model.layers.12.self_attn.o_proj	16,777,216	3,865,157,632
model.layers.12.self_attn.q_proj	16,777,216	3,881,934,848
model.layers.12.self_attn.v_proj	4,194,304	3,886,129,152
model.layers.13.input_layernorm	4,096	3,886,133,248
model.layers.13.mlp.down_proj	58,720,256	3,944,853,504
model.layers.13.mlp.gate_proj	58,720,256	4,003,573,760
model.layers.13.mlp.up_proj	58,720,256	4,062,294,016
model.layers.13.post_attention_layernorm	4,096	4,062,298,112
model.layers.13.self_attn.k_proj	4,194,304	4,066,492,416
model.layers.13.self_attn.o_proj	16,777,216	4,083,269,632
model.layers.13.self_attn.q_proj	16,777,216	4,100,046,848
model.layers.13.self_attn.v_proj	4,194,304	4,104,241,152
model.layers.14.input_layernorm	4,096	4,104,245,248
model.layers.14.mlp.down_proj	58,720,256	4,162,965,504
model.layers.14.mlp.gate_proj	58,720,256	4,221,685,760
model.layers.14.mlp.up_proj	58,720,256	4,280,406,016
model.layers.14.post_attention_layernorm	4,096	4,280,410,112
model.layers.14.self_attn.k_proj	4,194,304	4,284,604,416
model.layers.14.self_attn.o_proj	16,777,216	4,301,381,632
model.layers.14.self_attn.q_proj	16,777,216	4,318,158,848
model.layers.14.self_attn.v_proj	4,194,304	4,322,353,152
model.layers.15.input_layernorm	4,096	4,322,357,248
model.layers.15.mlp.down_proj	58,720,256	4,381,077,504
model.layers.15.mlp.gate_proj	58,720,256	4,439,797,760
model.layers.15.mlp.up_proj	58,720,256	4,498,518,016
model.layers.15.post_attention_layernorm	4,096	4,498,522,112
model.layers.15.self_attn.k_proj	4,194,304	4,502,716,416
model.layers.15.self_attn.o_proj	16,777,216	4,519,493,632
model.layers.15.self_attn.q_proj	16,777,216	4,536,270,848
model.layers.15.self_attn.v_proj	4,194,304	4,540,465,152
model.layers.16.input_layernorm	4,096	4,540,469,248
model.layers.16.mlp.down_proj	58,720,256	4,599,189,504
model.layers.16.mlp.gate_proj	58,720,256	4,657,909,760
model.layers.16.mlp.up_proj	58,720,256	4,716,630,016
model.layers.16.post_attention_layernorm	4,096	4,716,634,112
model.layers.16.self_attn.k_proj	4,194,304	4,720,828,416
model.layers.16.self_attn.o_proj	16,777,216	4,737,605,632
model.layers.16.self_attn.q_proj	16,777,216	4,754,382,848
model.layers.16.self_attn.v_proj	4,194,304	4,758,577,152
model.layers.17.input_layernorm	4,096	4,758,581,248
model.layers.17.mlp.down_proj	58,720,256	4,817,301,504
model.layers.17.mlp.gate_proj	58,720,256	4,876,021,760
model.layers.17.mlp.up_proj	58,720,256	4,934,742,016
model.layers.17.post_attention_layernorm	4,096	4,934,746,112
model.layers.17.self_attn.k_proj	4,194,304	4,938,940,416
model.layers.17.self_attn.o_proj	16,777,216	4,955,717,632
model.layers.17.self_attn.q_proj	16,777,216	4,972,494,848
model.layers.17.self_attn.v_proj	4,194,304	4,976,689,152
model.layers.18.input_layernorm	4,096	4,976,693,248
model.layers.18.mlp.down_proj	58,720,256	5,035,413,504
model.layers.18.mlp.gate_proj	58,720,256	5,094,133,760
model.layers.18.mlp.up_proj	58,720,256	5,152,854,016
model.layers.18.post_attention_layernorm	4,096	5,152,858,112
model.layers.18.self_attn.k_proj	4,194,304	5,157,052,416
model.layers.18.self_attn.o_proj	16,777,216	5,173,829,632
model.layers.18.self_attn.q_proj	16,777,216	5,190,606,848

continued on next page

Module	Params	Running total
model.layers.18.self_attn.v_proj	4,194,304	5,194,801,152
model.layers.19.input_layernorm	4,096	5,194,805,248
model.layers.19.mlp.down_proj	58,720,256	5,253,525,504
model.layers.19.mlp.gate_proj	58,720,256	5,312,245,760
model.layers.19.mlp.up_proj	58,720,256	5,370,966,016
model.layers.19.post_attention_layernorm	4,096	5,370,970,112
model.layers.19.self_attn.k_proj	4,194,304	5,375,164,416
model.layers.19.self_attn.o_proj	16,777,216	5,391,941,632
model.layers.19.self_attn.q_proj	16,777,216	5,408,718,848
model.layers.19.self_attn.v_proj	4,194,304	5,412,913,152
model.layers.20.input_layernorm	4,096	5,412,917,248
model.layers.20.mlp.down_proj	58,720,256	5,471,637,504
model.layers.20.mlp.gate_proj	58,720,256	5,530,357,760
model.layers.20.mlp.up_proj	58,720,256	5,589,078,016
model.layers.20.post_attention_layernorm	4,096	5,589,082,112
model.layers.20.self_attn.k_proj	4,194,304	5,593,276,416
model.layers.20.self_attn.o_proj	16,777,216	5,610,053,632
model.layers.20.self_attn.q_proj	16,777,216	5,626,830,848
model.layers.20.self_attn.v_proj	4,194,304	5,631,025,152
model.layers.21.input_layernorm	4,096	5,631,029,248
model.layers.21.mlp.down_proj	58,720,256	5,689,749,504
model.layers.21.mlp.gate_proj	58,720,256	5,748,469,760
model.layers.21.mlp.up_proj	58,720,256	5,807,190,016
model.layers.21.post_attention_layernorm	4,096	5,807,194,112
model.layers.21.self_attn.k_proj	4,194,304	5,811,388,416
model.layers.21.self_attn.o_proj	16,777,216	5,828,165,632
model.layers.21.self_attn.q_proj	16,777,216	5,844,942,848
model.layers.21.self_attn.v_proj	4,194,304	5,849,137,152
model.layers.22.input_layernorm	4,096	5,849,141,248
model.layers.22.mlp.down_proj	58,720,256	5,907,861,504
model.layers.22.mlp.gate_proj	58,720,256	5,966,581,760
model.layers.22.mlp.up_proj	58,720,256	6,025,302,016
model.layers.22.post_attention_layernorm	4,096	6,025,306,112
model.layers.22.self_attn.k_proj	4,194,304	6,029,500,416
model.layers.22.self_attn.o_proj	16,777,216	6,046,277,632
model.layers.22.self_attn.q_proj	16,777,216	6,063,054,848
model.layers.22.self_attn.v_proj	4,194,304	6,067,249,152
model.layers.23.input_layernorm	4,096	6,067,253,248
model.layers.23.mlp.down_proj	58,720,256	6,125,973,504
model.layers.23.mlp.gate_proj	58,720,256	6,184,693,760
model.layers.23.mlp.up_proj	58,720,256	6,243,414,016
model.layers.23.post_attention_layernorm	4,096	6,243,418,112
model.layers.23.self_attn.k_proj	4,194,304	6,247,612,416
model.layers.23.self_attn.o_proj	16,777,216	6,264,389,632
model.layers.23.self_attn.q_proj	16,777,216	6,281,166,848
model.layers.23.self_attn.v_proj	4,194,304	6,285,361,152
model.layers.24.input_layernorm	4,096	6,285,365,248
model.layers.24.mlp.down_proj	58,720,256	6,344,085,504
model.layers.24.mlp.gate_proj	58,720,256	6,402,805,760
model.layers.24.mlp.up_proj	58,720,256	6,461,526,016
model.layers.24.post_attention_layernorm	4,096	6,461,530,112
model.layers.24.self_attn.k_proj	4,194,304	6,465,724,416
model.layers.24.self_attn.o_proj	16,777,216	6,482,501,632
model.layers.24.self_attn.q_proj	16,777,216	6,499,278,848
model.layers.24.self_attn.v_proj	4,194,304	6,503,473,152
model.layers.25.input_layernorm	4,096	6,503,477,248
model.layers.25.mlp.down_proj	58,720,256	6,562,197,504
model.layers.25.mlp.gate_proj	58,720,256	6,620,917,760

continued on next page

Module	Params	Running total
model.layers.25.mlp.up_proj	58,720,256	6,679,638,016
model.layers.25.post_attention_layernorm	4,096	6,679,642,112
model.layers.25.self_attn.k_proj	4,194,304	6,683,836,416
model.layers.25.self_attn.o_proj	16,777,216	6,700,613,632
model.layers.25.self_attn.q_proj	16,777,216	6,717,390,848
model.layers.25.self_attn.v_proj	4,194,304	6,721,585,152
model.layers.26.input_layernorm	4,096	6,721,589,248
model.layers.26.mlp.down_proj	58,720,256	6,780,309,504
model.layers.26.mlp.gate_proj	58,720,256	6,839,029,760
model.layers.26.mlp.up_proj	58,720,256	6,897,750,016
model.layers.26.post_attention_layernorm	4,096	6,897,754,112
model.layers.26.self_attn.k_proj	4,194,304	6,901,948,416
model.layers.26.self_attn.o_proj	16,777,216	6,918,725,632
model.layers.26.self_attn.q_proj	16,777,216	6,935,502,848
model.layers.26.self_attn.v_proj	4,194,304	6,939,697,152
model.layers.27.input_layernorm	4,096	6,939,701,248
model.layers.27.mlp.down_proj	58,720,256	6,998,421,504
model.layers.27.mlp.gate_proj	58,720,256	7,057,141,760
model.layers.27.mlp.up_proj	58,720,256	7,115,862,016
model.layers.27.post_attention_layernorm	4,096	7,115,866,112
model.layers.27.self_attn.k_proj	4,194,304	7,120,060,416
model.layers.27.self_attn.o_proj	16,777,216	7,136,837,632
model.layers.27.self_attn.q_proj	16,777,216	7,153,614,848
model.layers.27.self_attn.v_proj	4,194,304	7,157,809,152
model.layers.28.input_layernorm	4,096	7,157,813,248
model.layers.28.mlp.down_proj	58,720,256	7,216,533,504
model.layers.28.mlp.gate_proj	58,720,256	7,275,253,760
model.layers.28.mlp.up_proj	58,720,256	7,333,974,016
model.layers.28.post_attention_layernorm	4,096	7,333,978,112
model.layers.28.self_attn.k_proj	4,194,304	7,338,172,416
model.layers.28.self_attn.o_proj	16,777,216	7,354,949,632
model.layers.28.self_attn.q_proj	16,777,216	7,371,726,848
model.layers.28.self_attn.v_proj	4,194,304	7,375,921,152
model.layers.29.input_layernorm	4,096	7,375,925,248
model.layers.29.mlp.down_proj	58,720,256	7,434,645,504
model.layers.29.mlp.gate_proj	58,720,256	7,493,365,760
model.layers.29.mlp.up_proj	58,720,256	7,552,086,016
model.layers.29.post_attention_layernorm	4,096	7,552,090,112
model.layers.29.self_attn.k_proj	4,194,304	7,556,284,416
model.layers.29.self_attn.o_proj	16,777,216	7,573,061,632
model.layers.29.self_attn.q_proj	16,777,216	7,589,838,848
model.layers.29.self_attn.v_proj	4,194,304	7,594,033,152
model.layers.30.input_layernorm	4,096	7,594,037,248
model.layers.30.mlp.down_proj	58,720,256	7,652,757,504
model.layers.30.mlp.gate_proj	58,720,256	7,711,477,760
model.layers.30.mlp.up_proj	58,720,256	7,770,198,016
model.layers.30.post_attention_layernorm	4,096	7,770,202,112
model.layers.30.self_attn.k_proj	4,194,304	7,774,396,416
model.layers.30.self_attn.o_proj	16,777,216	7,791,173,632
model.layers.30.self_attn.q_proj	16,777,216	7,807,950,848
model.layers.30.self_attn.v_proj	4,194,304	7,812,145,152
model.layers.31.input_layernorm	4,096	7,812,149,248
model.layers.31.mlp.down_proj	58,720,256	7,870,869,504
model.layers.31.mlp.gate_proj	58,720,256	7,929,589,760
model.layers.31.mlp.up_proj	58,720,256	7,988,310,016
model.layers.31.post_attention_layernorm	4,096	7,988,314,112
model.layers.31.self_attn.k_proj	4,194,304	7,992,508,416
model.layers.31.self_attn.o_proj	16,777,216	8,009,285,632

continued on next page

Module	Params	Running total
`model.layers.31.self_attn.q_proj`	16,777,216	8,026,062,848
`model.layers.31.self_attn.v_proj`	4,194,304	8,030,257,152
`model.norm`	4,096	8,030,261,248